Kethavath Prashanth

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.17
narayana junior college	Class XII (Telangana Board of Intermediate Education)	2024	97.20%
jawahar navodaya vidyalaya	Class X (central board of secondary education)	2022	84.40%

SKILLS

Languages: C, C++, Python, MySQL, CSS.

Computational Tools: Solid Edge, VHDL, Vesta, Quantum Espresso, Pymol.

Libraries: Numpy, Matplotlib, Pandas.

Developer Tools: Git, GitHub, Linux(Command Line), VS Code.

RELEVANT COURSES

Computer Science: Introduction to Programming, Data Structures and Applications, Operating Systems I & II

Scientific Computing: Computer Aided Numerical Methods I & II, Fundamentals of Scientific Computing, Computational Methods in Materials Science

Electrical: Digital Circuits, Introduction to HDL

Mathematics: Introduction to Probability, Introduction to Statistics, Linear Algebra, Transform Techniques, calculus I, calculs II, complex variables, differential equations.

PROJECTS

System-Call-Aware MLFQ Scheduler & Kernel Extension in xv6

Tech Stack**:** C, xv6 (RISC-V), QEMU, Operating Systems, Kernel Programming

- Extended the xv6 RISC-V operating system kernel by implementing custom system calls, process tracking mechanisms, and kernel-level statistics management.
- Designed and integrated a **4-level System-Call-Aware Multi-Level Feedback Queue (SC-MLFQ) scheduler** replacing the default round-robin scheduler with priority-based execution.
- Implemented process monitoring system calls including PID tracking, parent-child process analysis, syscall accounting, scheduling level retrieval, and MLFQ statistics reporting.
- Added adaptive scheduling features such as syscall-based interactive process detection, CPU-bound process demotion, time quantum management, and global priority boosting to prevent starvation.
- Developed user-space test programs on QEMU to validate system calls, scheduler correctness, process behavior, and workload performance.

Movie Recommendation System | Python

Tech Stack: Python, Pandas, NumPy, Scikit-learn, TF-IDF, Cosine Similarity, Streamlit, Machine Learning

- Developed a content-based movie recommendation system for 4,800+ movies using TF-IDF vectorization and cosine similarity.
- Engineered movie features from genres, keywords, cast, director, and plot overview to generate meaningful movie representations.
- Built an interactive Streamlit application with movie search and top-10 similarity-based recommendations.

Intelligent Search Engine with Autocomplete & Ranking

Tech Stack**:** C++, Trie, Hash Maps, LRU Cache, Algorithms, Data Structures

- Designed and developed an intelligent search engine supporting **prefix-based autocomplete** using the Trie data structure for efficient query retrieval.
- Implemented **ranking and recommendation algorithms** based on search frequency, recency, and user interaction to prioritize relevant search results.
- Built an **LRU Cache** to optimize repeated query performance and reduce search latency for frequently accessed results.
- Integrated **spell correction** using edit-distance-based matching to suggest relevant queries for misspelled inputs.

Molecular Dynamics Simulation Engine | Python, NumPy, Computational Physics

Tech Stack**:** Python, NumPy, Matplotlib, Computational Physics, Numerical Methods

- Built an atomistic MD simulator implementing Lennard-Jones interactions, PBC, FCC lattice generation, and Velocity Verlet numerical integration.
- Improved simulation performance using Verlet Neighbor Lists and vectorized force calculations.
- Analyzed phase behavior using RDF/MSD calculations, energy conservation, and diffusion properties.

POSITIONS OF RESPONSIBILITY

Undergraduate Mentor @ Sunshine

- Managed and Organized Multiple events such as Departmental Orientation, Mentor-Mentee Meets etc.
- Taking monthly mentorship sessions for the juniors on mental wellness,academic guidance impacting 30+ students.

Coordinator, events domine @ diesta

- inter department sports and cultural fest @IITH

Coordinator, logistics domine @ media council

- Coordinated logistics for institute events, managed resources, and collaborated with multiple teams to ensure smooth event execution.

EXTRACURRICULAR

- Silver Medal in power lifting, diesta Inter-Department Sports Event.
- Active participant in Cricket and Badminton, promoting teamwork, strategic thinking, and fitness.